Supplementary Information

Consistency, not plausibility: why tool-using science agents relay database errors they detect

George Xu

Supplementary Note 1: the first deployment run and its audit

The first full run of the deployment experiment began on 25 October 2026 under the author's own macOS account. Biomni's system prompt lists its data-lake directory, and that directory sat inside the study's repository, so the path shown to the model contained the strings `bioagent-plausibility` and `biomni_spike`. Biomni's code executor was not sandboxed: code the agent wrote could read the author's filesystem and reach the internet, and the study's repository was public.

After the run, every transcript was audited with three regular expressions, recorded in `results/analysis/audit` *awareness* (phrases inferring a plausibility benchmark, a test, or an evaluation of the agent), *repository pattern* (code executions referencing the author's home directory outside the data lake, the study's repository files, `git` commands or process listings) and *metadata vocabulary* (terms from the study's task files, such as `corrupted_value` or "answer key"). DeepSeek V4.1 Flash matched none of the first two in any run and the third in one run (u015), which caught the corruption regardless. Gemini 3.8 Flash matched awareness in 44 runs, the repository pattern in 98 and the vocabulary in 2. Awareness changed behaviour: Flash's aware runs relayed plausible values in 1 of 23 delivered runs, against 68 of 265 for the others, so the first run's Flash catch rate was inflated. The external files Flash fetched were Biomni's own repository (its evaluation task list) and a third-party benchmark (Bio-Bench); no transcript shows the study's files being read.

The corrupted conditions were rerun on 3 November 2026 under a second macOS user with no read access to the author's home directory, with the data lake at a neutral path (`/Users/Shared/agentrun/workspa` The agent's state file carried only the task identifier and a slot number; the mapping from slot to condition (`harness/rerun_schedule.json`, commit 0019d72) was read by the proxy and never shown to the agent. The clean runs were retained from the first run. In the rerun no transcript named the study, its repository or its paper. DeepSeek matched no pattern in any run. Flash matched awareness in 16 runs, each inferred from the task and the anomaly themselves rather than from the environment, the repository pattern in 17 runs of directory and process probing that found nothing, and the vocabulary in 2; the external files it fetched were again Biomni's own repository and Bio-Bench. Supplementary Table 1 gives both runs side by side; the main text reports the rerun throughout.

Supplementary Table 1: Deployment experiment, first run (October 2026) and rerun (3 November 2026). Relay is relayed/delivered runs; the odds ratio compares relay of plausible values on exact-route fields with other fields (registered test, task-clustered logistic regression, 95% interval). Audit columns count runs matching each pattern.

Model	Run	Impossible relayed	Plausible relayed	Odds ratio	Aware	Repository	Vocabular
DeepSeek V4.1 Flash	first	5/287 (2%)	80/285 (28%)	5.7 (3.1–10.4)	0	0	1
DeepSeek V4.1 Flash	rerun	4/293 (1%)	50/290 (17%)	17.2 (6.0–49.7)	0	0	0
Gemini 3.8 Flash	first	6/286 (2%)	69/288 (24%)	3.6 (2.0–6.5)	44	98	2
Gemini 3.8 Flash	rerun	4/283 (1%)	57/276 (21%)	6.9 (3.3–14.3)	16	17	2

Supplementary Table 2: Prompt variants: wrong value relayed (relayed/parsed runs) under the registered default prompt and three variants, on (a) protein tasks whose full record contradicts the value, (b) the chemistry agreement cell (three samples) and (c) coherent protein errors. The last column gives, for the two-fields variant, runs in which the stated database value matched the record (scored on (a) and (b)).

Cell	Model	Default	No tool instruction	Database has errors	Two fields	Stated value =
(a)	Opus 5	9/49 (18%)	4/49 (8%)	0/38 (0%)	5/47 (11%)	12/47 (26
(a)	Gemini 4 Argon	7/50 (14%)	6/50 (12%)	1/50 (2%)	0/50 (0%)	0/50 (0%
(a)	DeepSeek V4.1 Flash	50/50 (100%)	42/49 (86%)	23/42 (55%)	34/47 (72%)	13/47 (28
(b)	Opus 5	163/180 (91%)	85/180 (47%)	0/180 (0%)	0/180 (0%)	180/180 (1
(b)	Gemini 4 Argon	144/180 (80%)	92/180 (51%)	0/180 (0%)	0/180 (0%)	126/180 (7
(b)	Gemini 3.8 Flash	159/174 (91%)	58/171 (34%)	0/180 (0%)	6/163 (4%)	103/163 (6
(b)	DeepSeek V4.1 Flash	152/180 (84%)	140/180 (78%)	2/180 (1%)	19/180 (11%)	171/180 (9
(c)	Opus 5	49/50 (98%)	48/50 (96%)	50/50 (100%)	26/50 (52%)	–
(c)	Gemini 4 Argon	48/50 (96%)	47/50 (94%)	50/50 (100%)	0/50 (0%)	–
(c)	DeepSeek V4.1 Flash	47/50 (94%)	46/50 (92%)	49/49 (100%)	45/50 (90%)	–

Supplementary Table 3: Repetition versus consistency: wrong molecular weight relayed (relayed/parsed runs; 60 compounds, three samples, full record). R0, weight alone; R1, a verbatim repeat field scaled with the weight; R2, the repeat left correct; B, the heavy-atom count scaled with the weight. On uncorrupted records with the repeat field present, every model answered correctly in 60 of 60 runs. Pooled discordant pairs for RC1 (R0 relayed and R1 not, against the reverse): 198 against 84, two-sided McNemar $p = 10^{-11}$; for RC3 (R2 against R0), $p = 3 \times 10^{-144}$.

Model	R0 weight alone	R1 repeat agrees	R2 repeat disagrees	B derived field agrees
Opus 5	102/180 (57%)	66/180 (37%)	0/180 (0%)	113/180 (63%)
Gemini 4 Argon	117/180 (65%)	67/180 (37%)	3/180 (2%)	130/180 (72%)
Gemini 3.8 Flash	152/177 (86%)	135/178 (76%)	1/180 (1%)	154/175 (88%)
DeepSeek V4.1 Flash	115/180 (64%)	105/179 (59%)	5/180 (3%)	128/180 (71%)
Pooled	486/717 (68%)	373/717 (52%)	9/720 (1%)	525/715 (73%)

Supplementary Table 4: Memory probe on the 79 historical proteins. Memory state is the model's answer, without tools, for the protein's length: current (within one residue of today's length), superseded (within one of the superseded length), either (both, where they differ by at most two) or other. Relay of the superseded value is given as relayed/runs by memory state, without a second source (full record, one sample) and with the second source shown (annotate arm, three samples); p is the registered one-sided Fisher test of superseded-memory against current-memory relay with the second source.

Model	Memory state	Proteins	No second source	Second source shown	p
Opus 5	current	48	47/48	2/144	
	superseded	7	7/7	13/21	3×10^{-12}
	either	12	12/12	0/36	
	other	12	12/12	9/36	
Gemini 4 Argon	current	53	48/53	4/159	
	superseded	6	6/6	0/17	1
	either	12	11/12	1/36	
	other	8	6/8	0/24	
DeepSeek V4.1 Flash	current	50	50/50	38/150	
	superseded	6	6/6	17/18	1×10^{-8}
	either	11	11/11	7/33	
	other	12	11/12	18/36	

Supplementary Table 5: Withholding harness (full record, 100 tasks per domain and condition, one sample): corrupted numeric fields and identifiers relayed (relayed/parsed runs), runs without a parseable answer, and runs answered with a value that was neither the true nor the corrupted one. The last column gives identifier relay in the annotate arm of the benchmark (summary record) for comparison; the arms differ in record format.

Model	Domain	Condition	Numeric relayed	Identifiers relayed	Unparsed	Wrong	Annotate arm identi
Opus 5	proteins	impossible	0/46	0/50	4	0	0/100
		plausible	0/45	0/50	5	0	0/100
	molecules	impossible	0/75	5/25	0	0	21/25
		plausible	0/75	3/25	0	0	10/25
Gemini 4 Argon	proteins	impossible	0/49	0/50	1	0	0/100
		plausible	0/48	0/50	2	0	0/100
	molecules	impossible	0/75	4/25	0	0	19/25
		plausible	0/75	1/25	0	0	18/25
DeepSeek V4.1 Flash	proteins	impossible	0/34	4/50	16	5	1/100
		plausible	0/37	1/50	13	3	1/100
	molecules	impossible	0/75	12/25	0	0	22/24
		plausible	0/75	11/25	0	0	21/24